

Watering the Future: Assessing a Visualization Design for Accessible Comprehension of Water Management

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ABSTRACT

Managing water resources within California's Central Valley is a complex task involving analyzing a myriad of factors e.g. climate, demand, and government policies. This results in opaque management decisions which are concentrated to a small group of people while the majority of the valley is left out of the decision-making process, leading to inequitable distribution. This paper assesses the design for *Watering the Future*, an application aiming to educate non-experts in water management about water distribution systems by illustrating regional water supplies and the factors behind them in a more understandable manner. We sought feedback from a domain expert in water management and conducted a user study to assess the effectiveness of our graph designs, of which results suggested increased levels of engagement while still communicating the required information, suggesting an enhanced level of accessibility to the unfamiliar field of water management.

Index Terms: Visualization design study, public-facing visualization, visual metaphor

1 INTRODUCTION

Sharing a limited water supply is difficult in California's Central Valley, exacerbated by an uncertain climate future and conflicting water needs. Decisions in water allocation are centralized to a small group of decision-makers and are opaque to many residents due to the complexity and volume of water delivery data in the valley. This has provided momentum to increase transparency and democratize data related to water management decisions [7]. Due to its complexity, these decisions are assessed using a simulator called *CalSim3* [1] which generates water allocation data—time series of water delivery amounts to varying spatial locations.

Watering the Future is an online visualization and part of CO-EQWAL, a collaborative effort with domain experts in water management to develop a visualization design for water allocation data—a generally unfamiliar topic for non-experts in water supply management—with the main audience being residents of the Central Valley. We are motivated by the need to provide an alternative to the current standard visualization for such data, exceedance plots (Fig. 1b), which encode probability in the x-axis and water quantity in the y-axis [1]. While effective for domain experts, its design has been long identified in the community as challenging to the non-experts, including the notion of *exceedance probability* on which its design is based.

Designing an alternative that conveys exceedance probability while being approachable to non-experts is thus our main goal. In the language of Munzner's Nested Model for Visualization Design [16], this is our domain problem. Furthermore, these visualizations must

be made with the goal of decision-making in mind, where they must facilitate comparison tasks and illuminate causes behind particular water management cases.

The challenges we faced are similar to those of past works on designing visualizations for the public or in public spaces like museums [2,3,5,8,15,22]. Our challenge is primarily characterized by the duality of the need to design *engaging* visualizations for those with little knowledge or interest in water management data or strategies, while also ensuring that *complex* underlying data be communicated clearly. As documented by prior works outlining the visualization design process [16,19], developing visualizations in this constrained environment requires iterative feedback and design iterations, which we document for this project on a set of water allocation scenario data. In addition, we conduct expert reviews and a user study to assess our visualization design.

Our work contributes to the literature of designing public-facing visualizations of unfamiliar data with the principles of engagement and education in mind. Specifically, we document the use of unconventional visualization strategies to increase engagement with comparable results to industry-standard visualizations in understanding. Lastly, we reaffirm the need for continuous development integrating ample user and expert feedback to navigate the potential tradeoffs between these principles.

2 RELATED WORK

No past work to our knowledge specifically consider visualization of water allocation data; however, visualization design for the public has been the subject of many past works [5]. In particular, a body of past work outline considerations for visualization design specifically in a museum setting that bears resemblance to the communication setting for *Watering the Future* [3,8,15,21]. The similarities are mainly two-fold.

First, both cases assume that users cannot be expected to be familiar with the data or related concepts. Dasu et al., for example, consider the value of narrative strategies and animation to alleviate this knowledge gap [8]. For *Watering the Future*, we identified challenges with exceedance plots and collaboratively developed alternative visual encodings to support visual communication. Second, both cases require the visualization design to attract and engage users in the data presented, as the assumed level of initial interest to the data is low. Hornecker and Stifter reported that museum exhibits typically have ten seconds to incentivize viewers to interact with installments [11]. In such a setting, past works have identified the need to make the information visually appealing [2,10,21]. Of course, these considerations extend to web-based visualizations as well. Climate applications [5] to healthcare dashboards [23] also face similar constraints that require iterations to satisfy unique needs for public-facing visualizations where correctness and engagement are both paramount. The axes of memorability [6] and aesthetics [4] have also been considered in visualization design, as well as user engagement [12], informing our design process and evaluation.

3 WATERING THE FUTURE

To achieve the goal of facilitating decision-making, *Watering the Future* explains the concept of water distribution alternatives, called

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scenarios, and if/how they benefit various interest groups, called *objectives*, referred to as such due to its naming convention defining each group’s goals, e.g. the objective named "Folsom Storage" seeks to increase water storage allocation for the Folsom Dam.

3.1 Dataset

Factors such as yearly climate, an objective’s priority, government legislation, and demand affect water delivery quantities. A list of permutations of these factors (with each permutation thus representing a scenario) is pre-generated and fed into *CalSim* (Sec. 1), and the resulting deliveries are stored for later use. For *Watering the Future*, we use data generated by a lightweight variation of *CalSim* called *CalLite* [13] containing 11 objectives and 600 scenarios, resulting in a total of 6600 sets of water allocation data, each recorded in Thousand Acre-Feet (TAF) per year and spanning over 80 years.

3.2 Design Goals

Our process for *Watering the Future* followed established visualization design methodologies, particularly the nested model for visualization design by Munzner [16] and the design study methodology outlined by Sedlmair et al. [19]. In particular, several stages of our design closely follow several of the nine stages of Sedlmair et al. [19] which we will highlight. Having completed [19]—defined *Precondition Stages*—we moved to the *Discover* stage. We emphasized illustrating water delivery data such that a non-expert could understand while not simplifying the data to the point where its nuances, specifically probability, was removed. With feedback from a domain expert in water management, we defined the following design goals:

DG1: Communicate quantities and reliability of water delivery for a water allocation scenario. Exceedance plots convey quantity and probability, and the visualization must be able to communicate the same type of information.

DG2: Facilitate comparison between scenarios and between objectives. Water equity entails distribution plans which benefit as many people as possible. Thus, being able to compare deliveries across both the objective and scenario axes is a crucial feature.

DG3: Communicate causes of water delivery quantities and reliability. Each scenario is the product of changing various factors, so the visualization should be able to show the combination of factors contributing to a scenario.

4 VISUALIZATION DESIGN PROCESS

We outline our iterative design process in *phases*, continuing with the exploration of design alternatives for exceedance plots (*Design*).

4.1 Phase 1: Graph Designs

The quality of water deliveries can be represented most directly as a bar graph (Fig. 1a) where x-axis encodes time and y-axis encodes water delivery quantities. However, exceedance plots (Fig. 1b) are most often used by experts as they more effectively convey delivery probabilities. Notably, they omit a time axis due to the irrelevance of temporal progress and only communicate overall probabilities. As mentioned in Sec. 1, they are not familiar to non-experts. Bar graphs may be more familiar but not facilitate easy comparison of *probability* between delivery outcomes. Adjacent, we wanted to maintain the lack of a temporal axis to emphasize overarching probabilities and avoid distracting users with order of deliveries. Consistent with the nested model’s abstraction level [16], we explored how to transform probability data into representations accessible to non-experts, including a quantile dotplot and a novel bucket-like gradient glyph, inspired by gradient maps.

4.1.1 Gradient Glyphs

We designed gradient glyphs where water levels more likely to be exceeded are shaded darker, following a natural intuition where

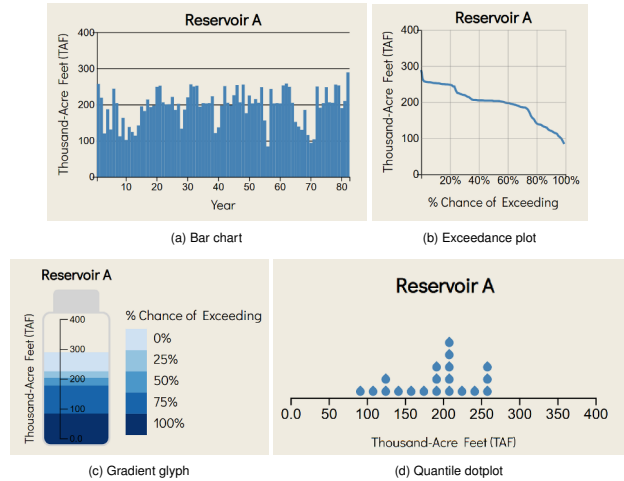


Figure 1: Four visualization designs for an example water allocation data labeled Reservoir A.

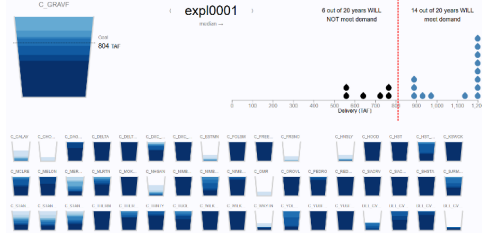


Figure 2: First interface prototype (Sec. 4.2) depicting every objective for scenario 0001 as a bucket-shaped gradient glyph. Objective labelled "C_GRAVF" is currently selected, shown as a large bucket at the top accompanied by its quantile dotplot.

surfaces more often submerged in water are stained darker due to repeated exposure to moisture. Thus, users can deduce that darker-colored water levels have a higher probability of being exceeded than lighter-colored water levels (DG1). Water level heights correspond to fixed percentiles of delivery data—every 10th percentile in the initial prototype and every 25th in the final design. We first developed these glyphs in a bucket shape with animations of water levels rising and falling in a wavelike motion to further communicate our water metaphor.

4.1.2 Dot Quantiles

Works in psychology identify that frequency framing of uncertain information may outperform probabilistic framing as in exceedance plots [9]. To explicitly operationalize frequency framing, we experimented with using quantile dotplots, a variation of histogram-like plots that performs well in conveying probabilities to end-users [14, 17] (DG1). Furthermore, we used droplet icons in place of dot icons to take advantage of visual metaphors and improve concept grasping.

4.2 Phase 2: First Interface Prototype

Our first interface (Fig. 2) listed one scenario across all objectives using bucket-shaped glyphs (DG2). Other scenarios are cycled through by clicking on buttons at the top of the interface, and the animations of each bucket would indicate the degree of change between water deliveries (DG2). Users could also choose to focus on one objective by clicking on its glyph which would reveal a dotplot of the current water allocation result and enable direct estimation of how many years a user can expect to meet a supply goal, adjustable via moving the razor indicated by a dotted red line (DG1). Users can thus decide what delivery scenarios satisfy a delivery goal for

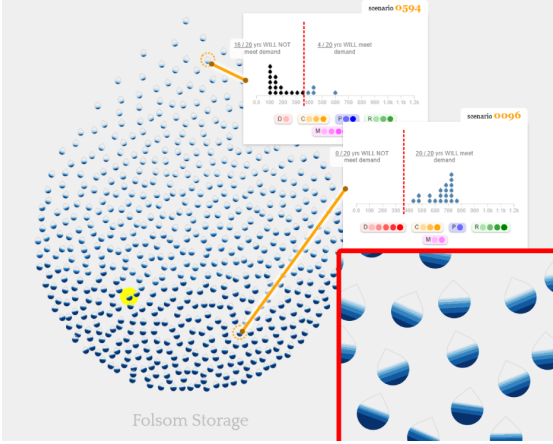


Figure 3: Examination view of the drill-down prototype (Sec. 4.3.1) with a zoomed-in frame outlined in red. The objective "Folsom Storage" is currently being examined with scenarios 0594 and 0096 selected.

their focused objective while seeing *other* buckets to decide what scenarios are *acceptable*, albeit non-ideal, that still allow others to receive water.

4.2.1 Feedback from Phase 2

We asked university students in a water resources and environment class to examine gradient glyphs and quantile dotplots and compare them to exceedance plots. Most agreed glyphs and dotplots were more intuitive to understand, but some preferred dotplots for comparison tasks, saying "the number of raindrops makes it a lot clearer what the ratio between [scenarios] is." Still, the students found the visual elements of the glyphs helped them conceptualize trade-offs between scenarios. These insights motivated the use of gradient glyphs and dotplots in future prototypes.

We also gathered feedback from our domain expert (Sec. 3.2). They remarked that the animations of the glyphs "add to the perception that it is a water bucket" and were instrumental in helping users quickly grasp the idea of water delivery data. Additionally, the expert commented that moving the slider in the quantile dotplot was compelling since users could interact with the data and see how it would apply to themselves, increasing interest and engagement. Thus, the animations and the razor were kept for future design iterations.

4.2.2 Design Decisions

When experimenting with larger datasets, we noticed the glyphs would overflow the screen (Fig. 2), overwhelming the user. To ameliorate this, we briefly tried modifying the glyphs to encode more information so *less* glyphs were needed on the screen by encoding scenario along the glyph's horizontal axis, but observed that further modifications to the glyph would compromise the water-level metaphor which was its core feature. Phase 2 marked the start of our **Implement** and **Deploy** loop

4.3 Phase 3: Drill-down Prototype

In line with the "overview first, zoom and filter, details on demand" mantra [20], we opted for a drill-down interface preceded by a tutorial for our second design. Since water droplets were already in our design language, we changed the glyph's shape to a waterdrop and grouped them into larger waterdrop groups to combat the scaling issue while simultaneously reinforcing the waterdrop visual language, providing continuity in the visualization design. The resulting interface is similar to Fig. 6a: glyphs belonging to the same objective are spatially arranged into a droplet shape, and these "droplet groups"

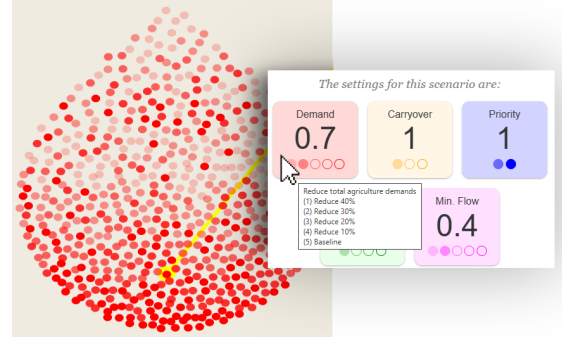


Figure 4: Supplementary note card in examination view (Sec. 4.3.1) in the final interface listing values of the factors for a selected scenario (not visible here). Saturation-encoded red circles represent the values of the hovered factor, demand, for all other scenarios.

are arranged into one waterdrop occupying the screen. Once a user selects a droplet group associated with an objective, they have the option to examine or compare the objective(s).

4.3.1 Examination View

If one objective is selected and the user clicks "examine", the screen changes to Fig. 3 which features all glyphs representing one objective ordered vertically by delivery mean. Here, users can hover or click their cursor over individual glyphs representing scenarios. A note card with a dotplot will then appear showing water delivery data for the scenario (DG1). Multiple scenarios may be selected to allow comparison (DG2).

This view also communicates *why* some scenarios have higher mean water deliveries (DG3). Each note card contains colored glyphs below the dotplot on which upon hovering, a second note card slides over presenting further information on the factors, similar to what is shown in Fig. 4. Each factor's numerical value occupies a square and outlines of small circles within directly correspond the factor's value, where an increased number of filled circles correlates to a higher value. Hovering over one square changes the waterdrop glyphs to saturation-encoded circles matching the color of the hovered square; in Fig. 4, the hovered red square is that of "demand". Users would notice that most saturated circles are lower on the waterdrop and infer that a higher demand largely correlates to higher deliveries. Thus, they may choose scenarios with higher demand to optimize for their objective.

4.3.2 Comparison View

If multiple objectives are selected, the "examine" option is replaced with a "compare" option and the screen changes to one similar to

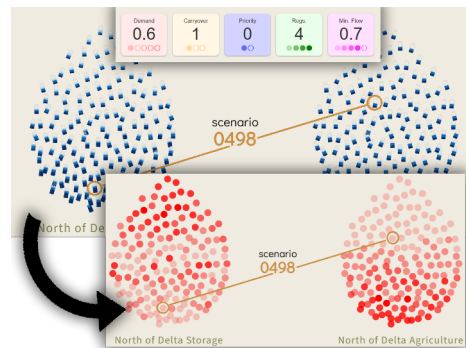


Figure 5: Comparison view (Sec. 4.3.2) in the final interface of the objectives labeled "North of Delta Storage" and "North of Delta Agriculture". Bottom picture depicts the view after hovering over the red square labeled "demand".

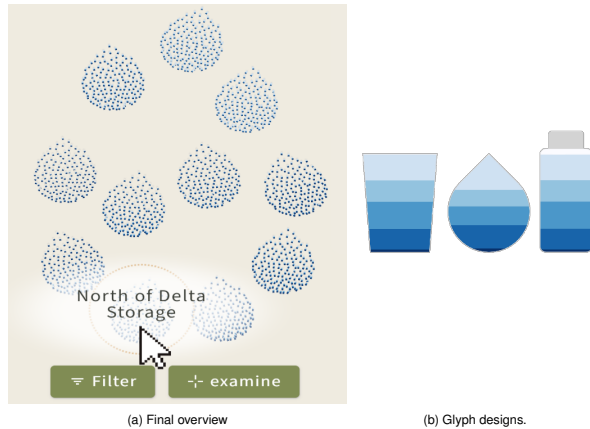


Figure 6: *Left*: Initial view of final interface (Sec. 4.4), with the objective labeled "North of Delta Storage" selected and hovered over. *Right*: Gradient glyphs in three shapes (left to right: bucket, droplet, bottle) with water quantities encoded by volume; the bottle shape is our final design.

Fig. 5 with the droplet groups adjacent. Most features remain the same as in Sec. 4.3.1, except only one scenario may be selected to compare how it performs across objectives; in Fig. 5, users would see scenario 498 is among the better alternatives for the objective "North of Delta Storage" (left group) since it is closer to the bottom of the droplet group than in "North of Delta Agriculture" (right). Further, the squares from Sec. 4.3.1 have moved to the top of the screen, and their interactions (which remain the same) reveal that higher demand results in *lower* deliveries for the Storage objective but *higher* ones for the Agriculture objective (DG3). Thus, both objectives may choose to compromise since choosing their respective "best" scenarios will be to the detriment of the other.

4.3.3 Expert Feedback from Phase 3 and Design Decisions

The expert reacted positively to the droplet spatial grouping, reflecting how it was "nice because you get to see all the possibilities in one screen." They were also impressed at the comparison view and highlighted its utility for users to view causes behind scenarios, benefiting users who are "interested in the what but also the why." One point of criticism was the irregularity of the glyph shape, remarking that users may be confused if deliveries were encoded by height or by volume—a discrepancy highlighted by Fig. 6b. Thus, we opted for a rectangular shape with our final water bottle glyph design.

4.4 Phase 4: Final Interface

Aside from replacing the glyph's droplet shape (Fig. 3), we also modified the color palette and fonts of the interface to those of the main COEQWAL website [7] to foster a more cohesive theme.

5 STUDY RESULTS AND DISCUSSION

To compare the efficacy between bottle-shaped glyphs and exceedance plots and assess DG2, we conducted an online survey where quantitative and qualitative data were obtained from 57 students in undergraduate psychology courses. (Reflect) [Further details in the Appendix.]

5.1 Results

Quantitative Results We created two tasks for each graph type: search for a water allocation dataset matching a set of measures, and find all measures matching a set of datasets. Thus, four sections of questions were presented in the survey, with the first task presented as multiple choice questions and the second presented as select all that apply (SATA) questions.

Time-to-insight (TTI) data measured in seconds showed broadly similar distributions and means across visualizations and question types: multiple choice: ($M_{EP} = 15.96$, $M_{BG} = 14.31$) and SATA ($M_{EP} = 49.49$, $M_{BG} = 53.6$). A linear mixed-effects model analysis found no significant differences in TTI between visualizations, suggesting comparable effects on performance. Post-test responses were collected through VisEngage, a 22-question 7-point Likert scale survey (7=highly agree) designed to assess user engagement along 11 characteristics [12]. Participants rated the gradient glyphs significantly above neutral on the characteristics "novelty", "attention", and "discovery", with 95% confidence intervals for both lying entirely above the midpoint of 4. One-sample t-tests were performed for all characteristics against a null hypothesis of a population mean of 4 ($t(56)$) at a significance level $\alpha = .05$. The results showed that the participants rated the gradient glyphs significantly higher than neutral in three characteristics: novelty, attention, and discovery. We found some evidence our glyph design outperformed in "aesthetic" and "exploration"; however, no other characteristics were suggestive of significance.

Qualitative Results After the the VisEngage survey, we asked participants to submit additional comments. One user described the bottle glyphs as "more entertaining and novel", noting that they helped communicate the meaning of exceedance more clearly once understood (DG3). Beyond novelty, one user suggested that the bottle glyphs helped to "convey the importance and application of the data". A common theme was that the concept of "exceedance" was difficult to comprehend. However, one user noted that "repeated exposure" made the tasks easier. In contrast, others preferred the exceedance plots, citing "familiarity with other graphs in classes" as a reason.

5.2 Expert Review

When our expert was asked if the tutorial section or any of the views supplied too much information, they commented "nothing is over-explained... if anything, you could add more [information]". They suggested adding text to assist users in interpreting the graphs, but acknowledged that one would need to "balance losing interest due to paragraphs of text". They also commented while the glyphs were now less confusing as bottle shapes, the overall visualization using a waterdrop shape for its spatial organization could make the user "think too much about what the... droplet means". This raises an issue of how modifying a visualization at the micro level could entail modifying the overall visualization—in this case, whether to sacrifice readability (glyph shapes) or an established visual language (waterdrop shape).

6 CONCLUSION

We document the design of *Watering the Future*, a visualization for non-experts on Central Valley water distribution, and assess our gradient glyph design for water allocation data. We draw on past work in designing public-facing visualizations, identified communication bottlenecks, and developed alternative designs with comparable performance but with better engagement metrics. Our design process aligns with several established visualization design methodologies [16, 19], although we only explicitly evaluated our process at what Munzner termed the encoding/interaction level. Further evaluation of the visualization system as a whole is thus a key future work. Our study reaffirms the value of exploring unconventional visualizations that depart from standard encodings to better engage public audiences. Through our design process, we repeatedly encountered a fundamental trade-off: visualizations optimized for precise information display often fail to engage non-expert users, while metaphorical designs that promote understanding may sacrifice some accuracy. This tension between precision and engagement, in addition to further design spaces for public-facing visualizations, warrants systematic investigation in future work.

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